April 29, 2026

SpiRIT: Psychology Appointment System Website

Group 4
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Abstract

- Web-based appointment management system
- Supports create, update, search, delete
- Uses Spring Boot + MariaDB + HTML/JS
- REST APIs + AJAX for communication
- Improves efficiency and reduces errors

Problem Definition

- Manual systems cause scheduling conflicts
- No real-time visibility
- Difficult to access/update records
- Poor data management
- No proper feedback system

Solution

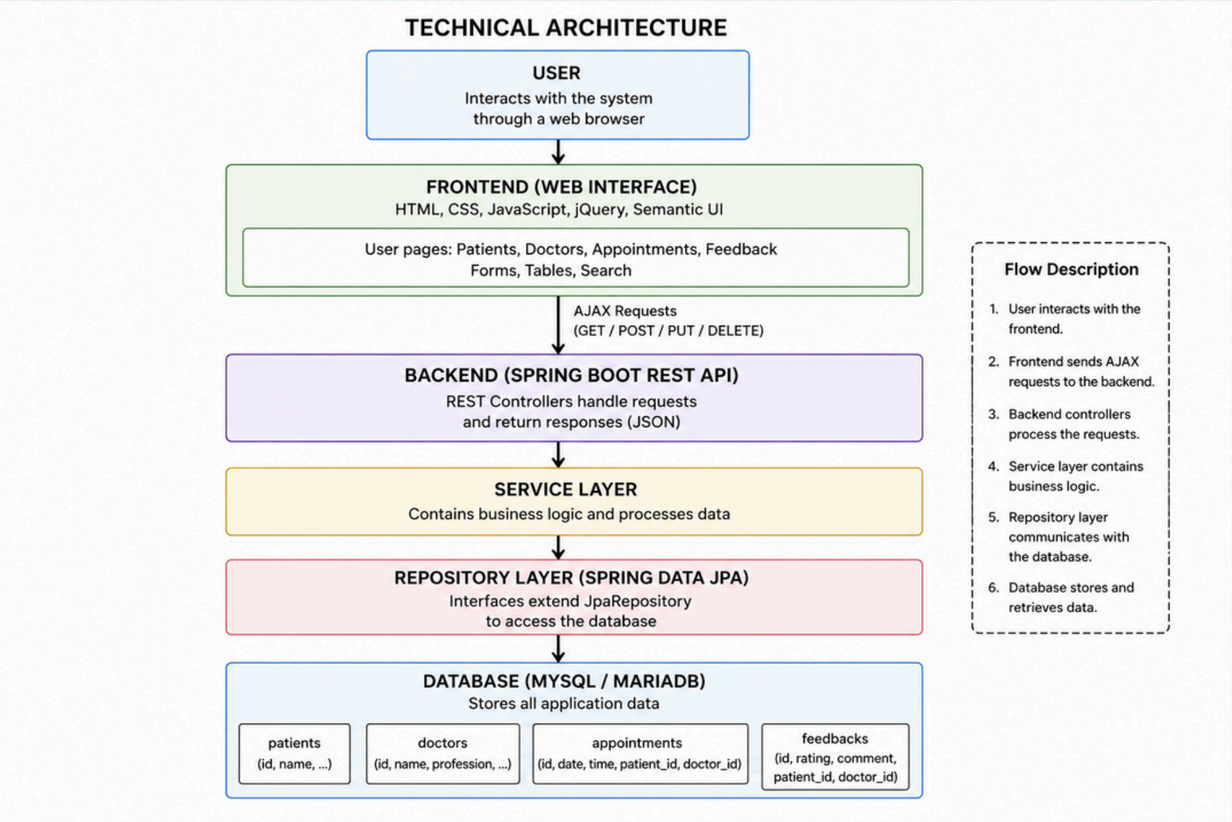
- Web-based centralized system
- Uses database + REST APIs
- Secure and structured data storage
- Easy appointment scheduling
- Feedback collection for improvement

Technologies Used

- Spring Boot (Backend)
- HTML, CSS, JavaScript
- jQuery (AJAX)
- Semantic UI
- MariaDB (Database)

Technical Architecture

- Layered architecture
- Frontend → HTML, CSS, JS
- AJAX sends requests
- REST Controllers handle requests
- Services process logic
- Repositories access database



Database Design

Entities	Relationships
Patient	Patient → Appointments
Doctor	Doctor → Appointments
Appointment	Patient → Feedback
Feedback	Doctor → Feedback

Features

Patient

Add, update, delete, search

Doctor

Add, update, delete, Search by name/profession

Appointment

Create, update, delete, Search by department

Feedback

Create, update, delete, Rating, comment, Search by multiple fields

API Overview

Patient

- GET /api/patients: Retrieve all patients
- GET /api/patients/{id}: Retrieve patient by ID
- GET /api/patients/search?name={name}: Search patients by name
- POST /api/patients: Create a new patient
- PUT /api/patients/{id}: Update a patient
- DELETE /api/patients/{id}: Delete a patient

Doctor

- GET /api/doctors: Retrieve all doctors
- GET /api/doctors/{id}: Retrieve doctor by ID
- GET /api/doctors/search?name={name}: Search doctors by name
- GET /api/doctors/search?profession={profession}: Search doctors by profession
- POST /api/doctors: Create a new doctor
- PUT /api/doctors/{id}: Update a doctor
- DELETE /api/doctors/{id}: Delete a doctor
- PATCH /api/doctors/{id}/profession: Update only doctor profession

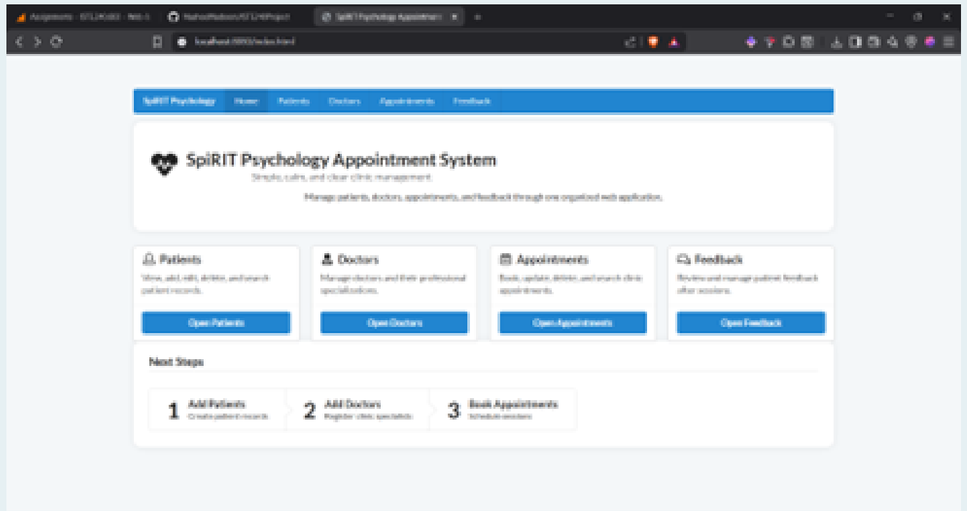
Feedback

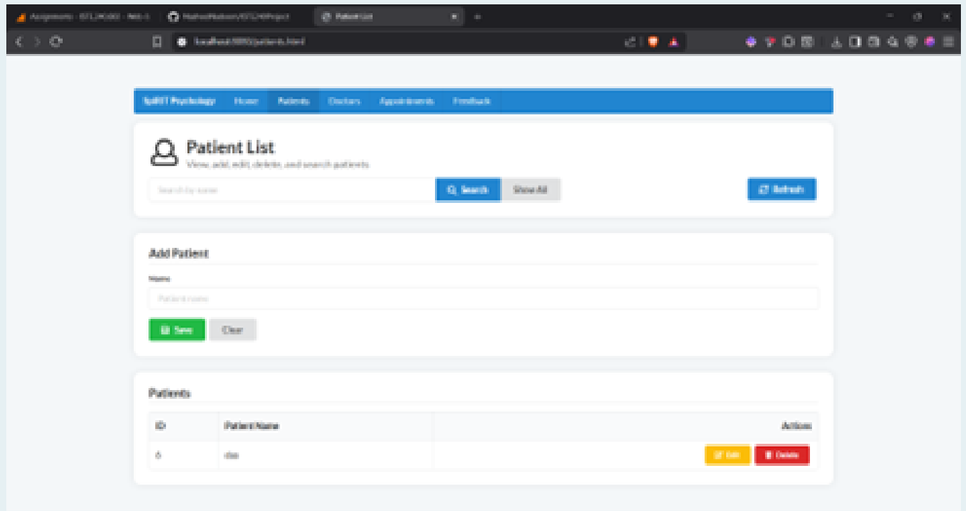
- GET /api/feedbacks: Retrieve all feedback
- GET /api/feedbacks/{id}: Retrieve feedback by ID
- GET /api/feedbacks/search/doctor?name={name}: Search feedback by doctor name
- GET /api/feedbacks/search/patient?name={name}: Search feedback by patient name
- GET /api/feedbacks/search/rating?rating={rating}:
- Search feedback by rating
- GET /api/feedbacks/search/comment?keyword={keyword}: Search feedback by comment
- POST /api/feedbacks:
- Create new feedback
- PUT /api/feedbacks/{id}:
- Update feedback
- DELETE /api/feedbacks/{id}:
- Delete feedback

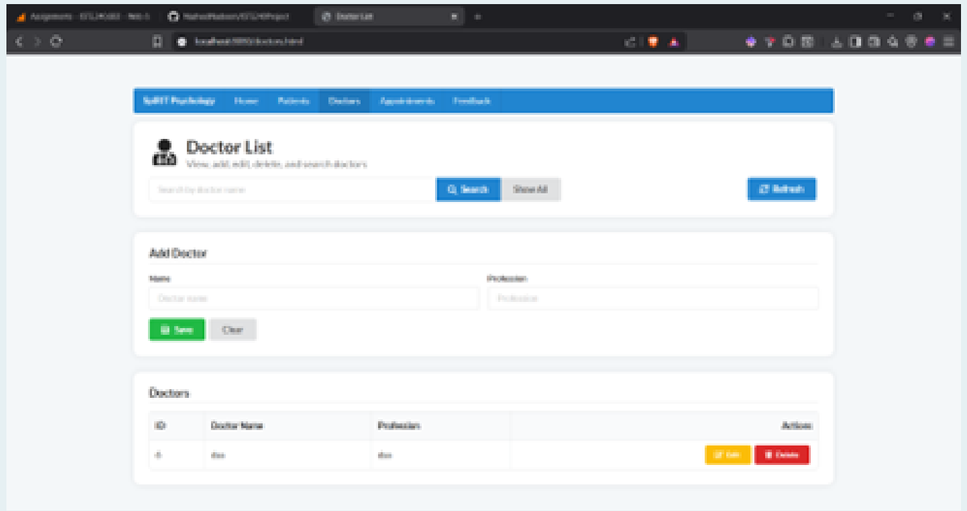
Appointment

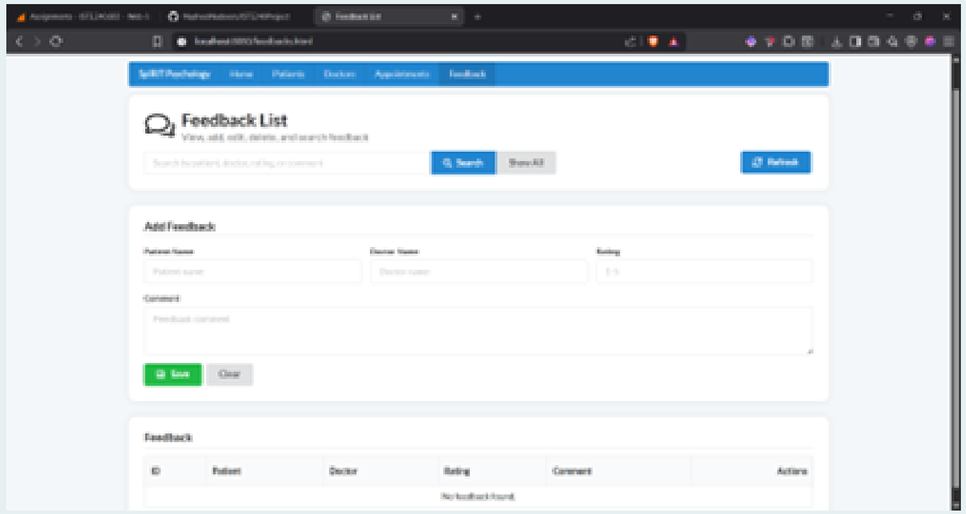
- GET /api/appointments: Retrieve all appointments
- GET /api/appointments/{id}: Retrieve appointment by ID
- GET /api/appointments/search?department={department}: Search appointments by department
- POST /api/appointments: Create a new appointment
- PUT /api/appointments/{id}: Update an existing appointment
- DELETE /api/appointments/{id}: Delete an appointment
-

Frontend examples









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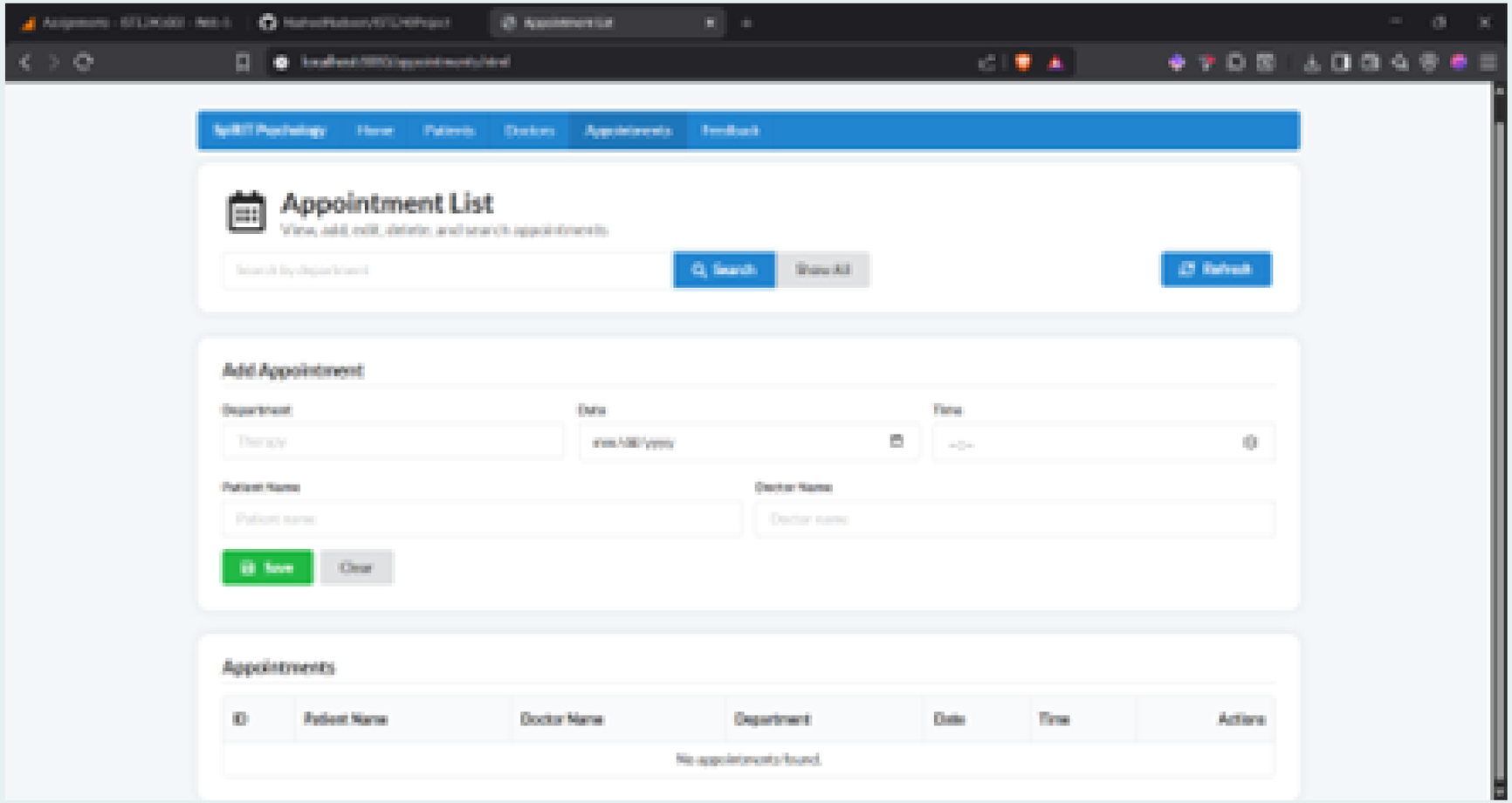

Demo

GitHub & Collaboration

Each student used separate branch
Used commits + pull requests
Merged into main branch

Team Contributions

Mohammed Ahmad

Appointment Module, Service, Repository, Controller and HTML page.
Managing Github merging.
Reported on Database Design and Features, and the API documentation.

Mohamed Ahmed

Feedback Module, Service, Repository, Controller and HTML page.
Contributed in writing the abstract.

Nasheel Nadeem

Doctor Module, Service, Repository, Controller and HTML page.
Contributed in the writing of Similar solutions, Technologies used and Technical Architcture.

Ryan Benson

Patient Module, Service, Repository, Controller and HTML page.
Wrote problem definition.

Questions